In [1]:
```
from pathlib import Path
import pandas as pd
import numpy as np
DATA_PATH = '/content/bookbazaar_orders_2026.csv';
df = pd.read_csv(DATA_PATH)
print('Rows and columns:', df.shape)
print(df.head(3).to_string(index=False))
```

```
Rows and columns: (86, 19)
Order_ID Customer_ID Customer_Name Product_ID              Product   Category
Brand  Quantity Unit_Price  Discount_Percent  Shipping_Cost Payment_Method Order_
Date  Ship_Date Delivery_Status        City Region  Rating Return_Status
 ORD0001        C212  Customer 212       B001 Python for Beginners Technology  Br
ightInk         2        799              10.0           60.0    Credit Card 2026
-01-17 2026-01-19       Delivered Bhubaneswar   East     4.0            No
 ORD0002        C216  Customer 216       B012  Adventure Storybook   Children Lit
tleLeaf         1        249              15.0           40.0            UPI 2026
-01-07 2026-01-12       Delivered     Chennai  South     4.0            No
 ORD0003        C221  Customer 221       B009         The Lost Map    Fiction  Pa
geCraft         1        349              10.0            0.0    Credit Card 2026
-01-08 2026-01-13       Delivered     Chennai  South     4.0            No
```

In [2]:
```
# Missing values
print('Missing values in every column:')
print(df.isnull().sum())
```

```
Missing values in every column:
Order_ID            0
Customer_ID         0
Customer_Name       1
Product_ID          0
Product             0
Category            0
Brand               0
Quantity            0
Unit_Price          0
Discount_Percent    1
Shipping_Cost       1
Payment_Method      0
Order_Date          0
Ship_Date           2
Delivery_Status     0
City                0
Region              0
Rating              1
Return_Status       0
dtype: int64
```

In [3]:
```
# Columns with missing values
print('Columns containing missing values:')
print(df.isnull().sum()[df.isnull().sum() > 0])
```

```
Columns containing missing values:
Customer_Name       1
Discount_Percent    1
Shipping_Cost       1
Ship_Date           2
Rating              1
dtype: int64
```

In [4]:
```
print('Total missing values:')
print(df.isnull().sum().sum())
```

```
Total missing values:
6
```

In [5]:
```
print("Rows containing missing values:")
print(df[df.isnull().any(axis=1)])
```

```
Rows containing missing values:
   Order_ID Customer_ID Customer_Name Product_ID               Product  \
7   ORD0008        C231           NaN       B008   Build Better Habits
18  ORD0019        C204  Customer 204       B008   Build Better Habits
27  ORD0028        C224  Customer 224       B008   Build Better Habits
39  ORD0040        C204  Customer 204       B010      Mountain Mystery
41  ORD0042        C223  Customer 223       B001  Python for Beginners
58  ORD0059        C226  Customer 226       B010      Mountain Mystery

      Category          Brand  Quantity Unit_Price  Discount_Percent  \
7    Self-Help  Sunrise Books         1        399              15.0
18   Self-Help  Sunrise Books         3        399               NaN
27   Self-Help  Sunrise Books         3        399               0.0
39     Fiction      NorthStar         1        299               0.0
41  Technology      BrightInk         3        799               5.0
58     Fiction      NorthStar         1        299               0.0

    Shipping_Cost Payment_Method  Order_Date   Ship_Date Delivery_Status  \
7             0.0    Credit Card  2026-01-25  2026-01-27       Delivered
18           40.0    Credit Card  2026-02-11  2026-02-13       Delivered
27            NaN     Debit Card  2026-02-01  2026-02-06       Delivered
39           30.0            UPI  2026-03-26  2026-03-30        Returned
41           40.0     Debit Card  2026-03-21         NaN       Cancelled
58           40.0    Credit Card  2026-05-14         NaN       Cancelled

           City Region  Rating Return_Status
7       Chennai  South     4.0            No
18    Bengaluru  South     3.0            No
27    Bengaluru  South     5.0            No
39         Pune   West     NaN           Yes
41  Bhubaneswar   East     4.0            No
58         Pune   West     5.0            No
```

In [6]:
```
df["Customer_Name"] = df["Customer_Name"].fillna("Unknown")
print("Missing values after filling Customer_Name:")
print(df.isnull().sum()[df.isnull().sum() > 0])
```

```
Missing values after filling Customer_Name:
Discount_Percent    1
Shipping_Cost       1
Ship_Date           2
Rating              1
dtype: int64
```